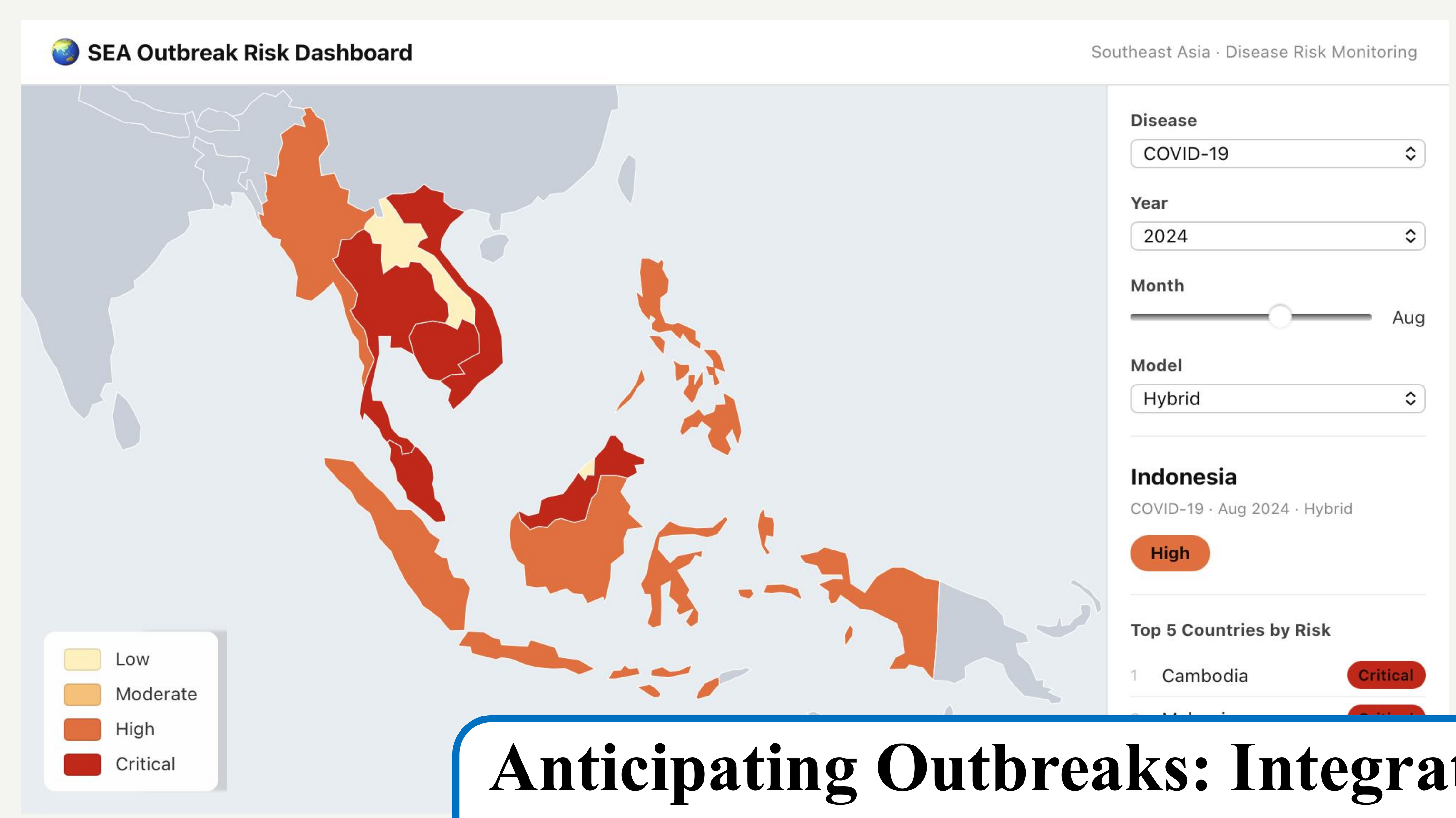

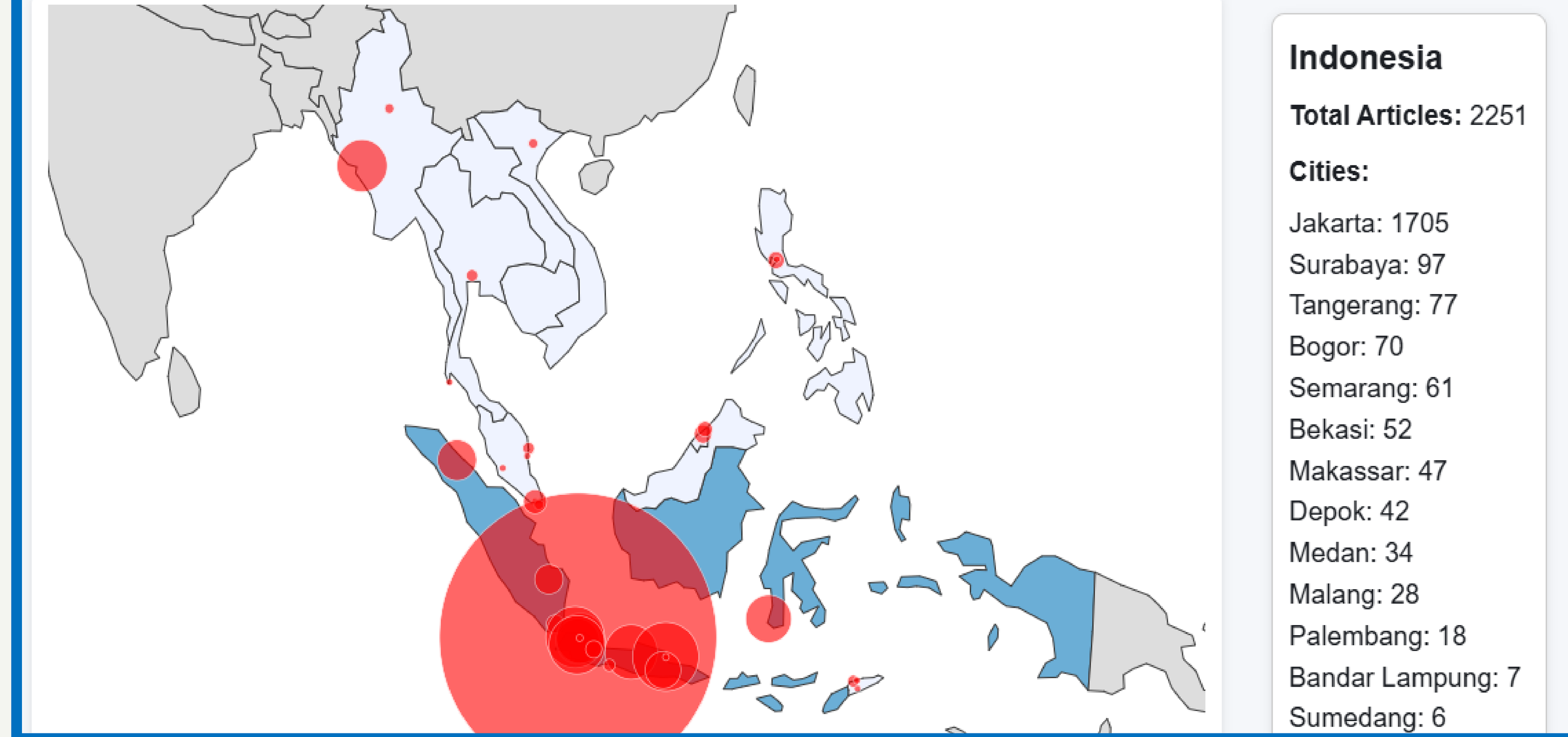

Anticipating Outbreaks: Integrating Multi-Source Signals for Early Detection and Decision Support

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MOTIVATION

What is the problem?

- Disease outbreaks are often detected too late to prevent widespread harm
- Early warning signs exist, but they are scattered across news, online behavior, and supply disruptions
- The Association of Southeast Asian Nations (ASEAN) often lack the resources and infrastructure available in wealthier global counterparts to monitor these signals quickly

Why is this important?

- ASEAN is home to hundreds of millions of people who are more vulnerable to delayed health responses
- Under-resourced health systems are overwhelmed quickly once outbreaks spread (e.g., COVID-19), and depend on early detection to mitigate escalation
- Providing timely, accessible health intelligence promotes equity and protects populations most at risk

APPROACHES

Our Approach

- A multi-source early-warning algorithm to detect outbreak risk before official case reports
- An interactive dashboard that converts model outputs into risk insights

Algorithm

- Combine official outbreak reports with faster, informal signals:
  - World Health Organization (WHO) epidemiological reports
  - Google search activity and news coverage
  - International Monetary Fund (IMF) supply chain and mobility patterns
- Train a supervised model using WHO news case and death data
- Classify monthly risk levels relative to each country's baseline

Visualization

- Weekly ASEAN country-level dashboard showing:
  - Current risk level and trends over time
  - Deviations from recent baseline
- Users can drill down to identify which data sources and signals drive risk

Why does this work?

- Outbreaks generate signals across many systems before lab confirmation
- Informal data often captures early weak signals
- Cross-source triangulation reduces noise and false alarms
- Earlier detection enables faster investigation and response

What's new?

- Moves beyond single-source surveillance by jointly modeling multiple leading indicators
- Uses country-specific baselines, improving relevance across diverse ASEAN settings
- Integration prediction, interpretation, and decision-support in one system
- Focus on practical early warning, not reactive analysis

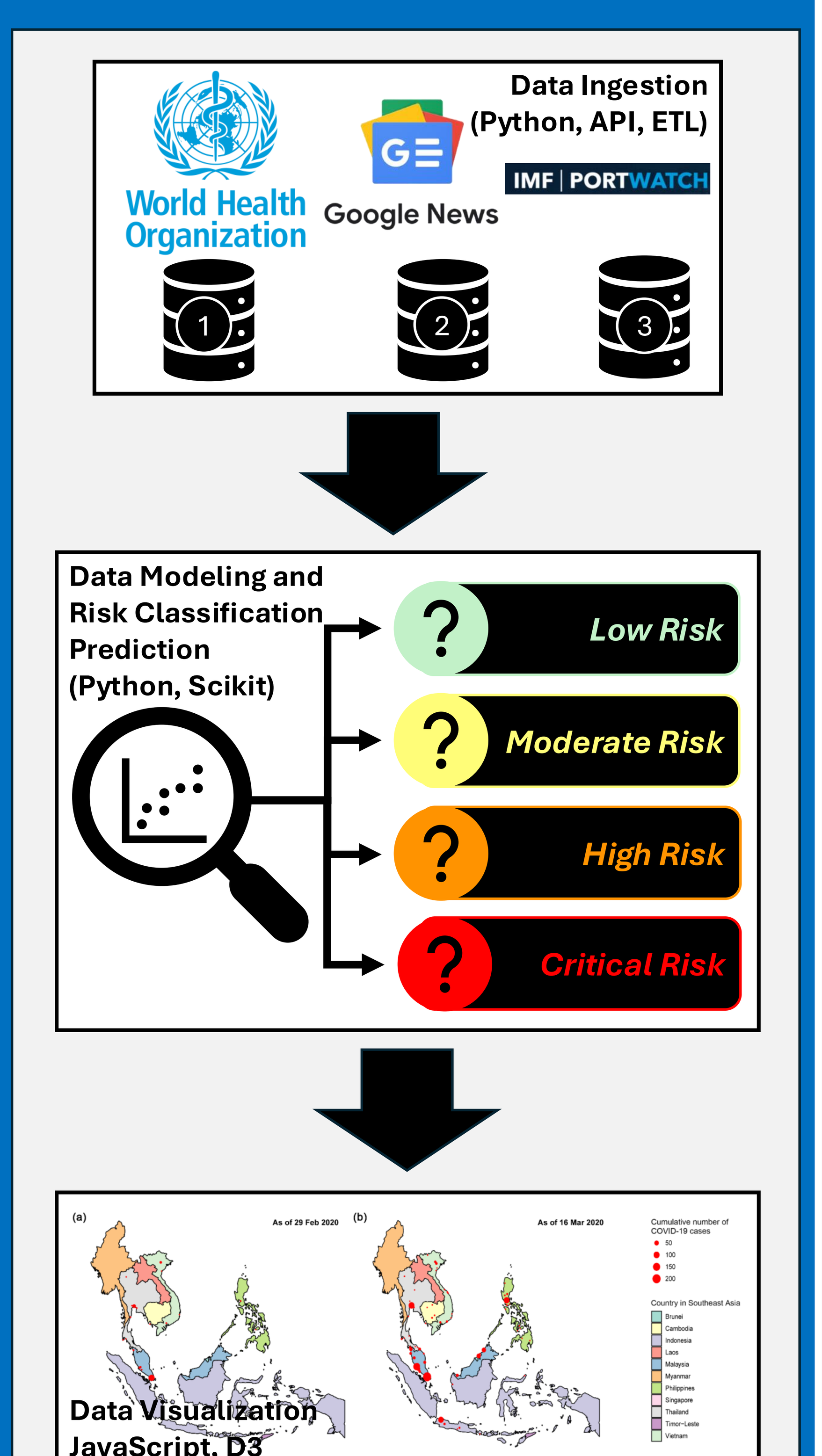

Figure 1. End-to-End Pandemic Risk Prediction Pipeline

Data

Type	Source	Method	Traits	Key Variables
Outbreak reports, demographics	WHO, UN	Manual data file download	114 KB, 3168 records, 38 features	ISO week; country; disease; case and death counts; linkable identifiers for matching to signals; mobility_index; ihr_score; beds_per_10k; physicians_per_10k; pop_density; urban_pct; temp; humidity
News, social media, and internet search behavior	Online news outlets, public posts, Google Trends	API – GDELT, web scraping	8.7 MB, 2579 records, 86 features	Article timestamp and ISO week; headline/body text; location mentions; disease/symptom keywords; count of relevant articles; and sentiment; post timestamp and ISO week; text content; keywords; inferred/declared location; counts of symptom/disease mentions per place-time window; search interest index by keyword/topic; time series at daily/weekly resolution; curated symptom/disease keyword lists
Logistics & mobility	World Bank and IMF	Manual data file download, API – IMF	100 MB, 262k records, 57 features	Goods and services volume trades; deviations from baseline mobility; connectivity features (in-/out-flow).

Figure 2. Overview of Data Sources, Collection Methods, and Key Variables Used for Pandemic Risk Modeling

Results

Evaluation Methods

- Four classification models were developed to predict historical ASEAN pandemic risk for each country-month-disease combination:
  - Health: outbreak reports, demographic baseline features
  - Media: news, social media, and internet search signals
  - Logistics: supply chain, mobility, and logistics indicators
  - Hybrid: uses the top 10 features from the health, media, and logistics models
- Due to data availability, evaluation focused on:
  - 10 ASEAN countries
  - 3 diseases: Avian Influenza, Dengue, and COVID-19
  - 96 months: Jan 2018 to Dec 2025
  - Actual risk labels were adjusted using available WHO Disease Outbreak News reports and outbreak severity.
    - $Monthly\ Severity\ Score(country, month, disease) = \log(1 + reported\ cases) + 2\log(1 + deaths)$
      - Deaths were weighted more heavily than reported cases to better reflect outbreak severity
      - For each country-disease pair, monthly severity scores were ranked and converted into four ordered labels:
      - $$Risk\ Label(country, month, disease) = \begin{cases} Low & \\ Moderate & \\ High & \\ Critical & \end{cases} if \begin{array}{l} 1^{st} \le Percentile_{Sev\ Score} \le 25^{th} \\ 25^{th} < Percentile_{Sev\ Score} \le 50^{th} \\ 50^{th} < Percentile_{Sev\ Score} \le 75^{th} \\ 75^{th} < Percentile_{Sev\ Score} \le 99^{th} \end{array}$$
- Model performance was evaluating using:
  - AVG_ADRL: average absolute difference between predicted and actual ordered risk labels, where lower values indicate closer predictions (e.g., low – moderate = 1, critical – moderate = 2)
  - Match Rate: proportion of predictions that exactly matched the adjusted actual risk label

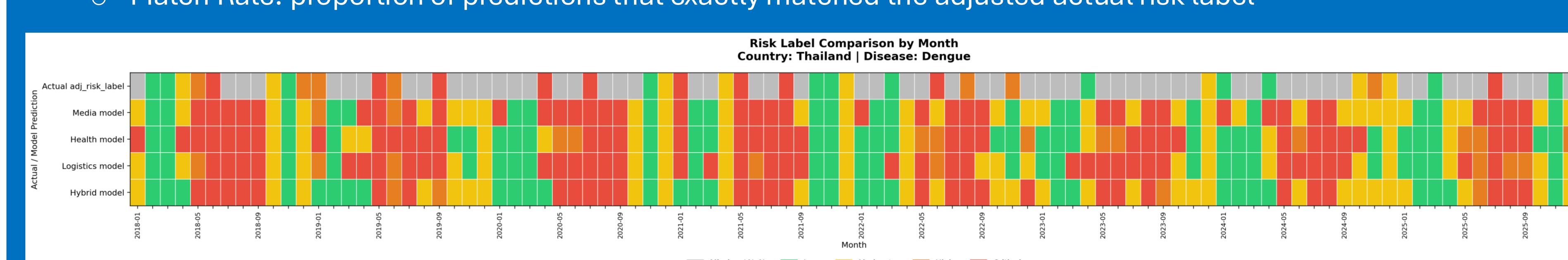

Figure 3. Monthly Risk Label Comparison Across Actual and Model-Predicted results for Dengue in Thailand

Summary of Results

- Logistics performed best overall, with the lowest AVG_ADRL 0.65 and highest Match Rate 0.58.
- Media/Web was highly competitive, especially for Dengue, where it achieved the best disease-level AVG_ADRL 0.53.
- Dengue was the easiest disease to predict overall, with AVG_ADRL 0.60, while COVID-19 was hardest, with AVG_ADRL 0.90.
- Country performance varied widely: Thailand performed best with AVG_ADRL 0.55, while the Philippines had the highest error at 0.87.
- The Hybrid model did not outperform individual models, suggesting that combining all data sources introduced noise and that source-specific models may be more reliable.

Figure 4. Overall Model Performance Comparison

Model	Best Algorithm	Sample Size	AVG_ ADRL	Match Rate
Health (Epi)	Logistic Classifier	562	0.71	0.55
Media (Web)	Extra Trees Classifier	479	0.68	0.57
Logistics (Log)	Random Forest	479	0.65	0.58
Hybrid (Hyb)	Random Forest	479	0.83	0.48

Figure 5. Overall Model Performance Comparison – By Disease

Disease	Sample Size	AVG_ ADRL	EPI_AVG_ ADRL	WEB_AVG_ ADRL	LOG_AVG_ ADRL	HYB_AVG_ ADRL
Dengue	1031	0.60	0.65	0.53	0.55	0.65
Avian Influenza	228	0.65	0.63	0.60	0.54	0.85
COVID-19	740	0.90	0.81	0.92	0.81	1.08

Visualization Insights

- How to Use Dashboard:
  1) Select a disease and year to view historical outbreak risk
  2) Click a month and country to see trends and the signals driving risk
  3) Switch models to compare which data sources matter most

Figure 6. Overall Model Performance Comparison – By ASEAN Country

Country	Sample Size	AVG_ ADRL	EPI_AVG_ ADRL	WEB_AVG_ ADRL	LOG_AVG_ ADRL	HYB_AVG_ ADRL
Thailand	208	0.55	0.58	0.56	0.5	0.56
Lao PDR	39	0.56	0.56	N/A	N/A	N/A
Cambodia	228	0.62	0.60	0.70	0.51	0.68
Myanmar	304	0.71	0.79	0.66	0.59	0.79
Singapore	212	0.71	0.79	0.49	0.58	0.96
Vietnam	284	0.72	0.62	0.72	0.69	0.85
Brunei	44	0.73	0.73	N/A	N/A	N/A
Malaysia	168	0.74	0.52	0.64	0.79	1.00
Indonesia	232	0.82	0.88	0.79	0.76	0.84
Philippines	280	0.87	0.87	0.86	0.77	0.97

Methods Comparison

- Traditional WHO DON and regional health reports are valuable, but they can be sporadic, retrospective, and unavailable for many country–disease–month combinations.
- Our approach uses routine multi-source signals from health, media/web, logistics, and hybrid features to estimate pandemic risk when official reporting is limited.
- The dashboard and heatmaps move beyond static tables by making risk patterns easier to compare across countries, diseases, months, and models.
- Using non-health signals such as news coverage, search behavior, mobility, and logistics activity expands situational awareness beyond traditional surveillance data.
- This method supports more real-time, up-to-date, and interpretable outbreak monitoring, especially for vulnerable populations in areas with less frequent official reporting.

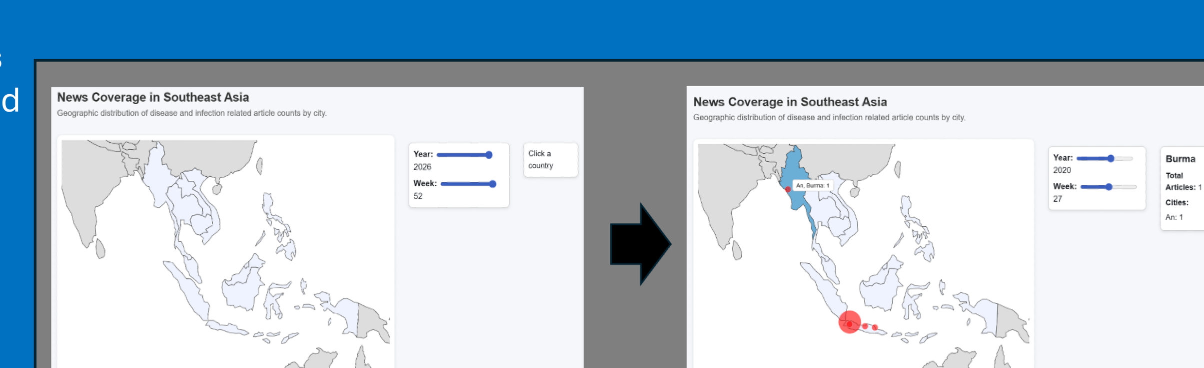

Figure 7. Dashboard – News Coverage Demonstration